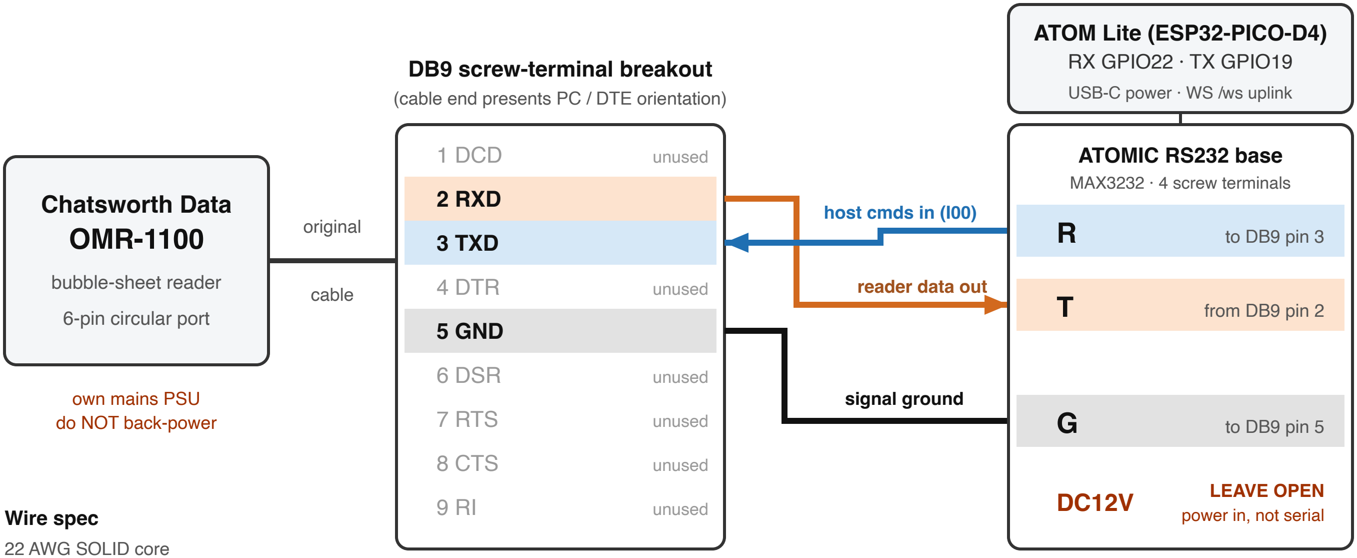
OMR-1100 RS-232 tap — wiring schematic

Three wires. DB9 screw-terminal breakout into the M5Stack ATOMIC RS232 base (MAX3232). Verified end-to-end 2026-07-30 — two cards decoded.



Wire spec

22 AWG SOLID core
15–20 cm, strip 5–6 mm
black for GND
stranded frays out of the
clamp — tin or ferrule it

Hazards

- DC12V is the base's power input, NOT a serial line. Nothing from the reader touches it. Power the ATOM over USB-C.
- NEVER wire the reader's TX straight to a bare ATOM GPIO. RS-232 swings ±5–12 V and will fry the 3.3 V pin. It must pass through the MAX3232 — the screw terminal is on its RS-232 side.
- Do not back-power the reader; it has its own external mains PSU.

Why only three

- No handshake lines. This unit's EEPROM flags byte reads 00 — flow control off, so DCD/DTR/DSR/RTS/CTS/RI all stay unconnected.
- The link is BIDIRECTIONAL — T is not optional. The host must send the I00 mode download or the reader emits nothing at all.
- **THE LABELS CROSS. Pin 2 goes to T and pin 3 to R — the base names** its terminals for the FAR device, not for the ESP32. Wired the intuitive way (2→R, 3→T) the link is totally silent.